DER Models

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1 Without Storage

1.1 Disaggregation

$$
\begin{aligned}
\max \quad & \sum_{t \in T} \left(P_t^{DA} x_{it} + \mathbb{E}\left[P_t^{RT}(\xi) y_{it}^{+}(\xi) - P_t^{PN} y_{it}^{-}(\xi) \right] \right) && \text{(1a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) \quad \forall t \in T && \text{(1b)} \\
& R_{it}(\xi) \geqslant y_{it}^{+}(\xi) \quad \forall t \in T && \text{(1c)} \\
& y_{it}^{+}(\xi) \leqslant M z_{it}(\xi), \quad y_{it}^{-}(\xi) \leqslant M(1 - z_{it}(\xi)) \quad \forall t \in T && \text{(1d)} \\
& x_{it}^{DA} \geqslant 0, y_{it}^{+}(\xi) \geqslant 0, y_{it}^{-}(\xi) \geqslant 0, z_{it}(\xi) \in \{0, 1\} \quad \forall t \in T && \text{(1e)}
\end{aligned}
$$

1.2 Aggregation

$$
\begin{aligned}
\max \quad & \sum_{t \in T} \left(P_t^{DA} \alpha_t + \mathbb{E}\left[P_t^{RT}(\xi) \beta_t^{+}(\xi) - P_t^{PN} \beta_t^{-}(\xi) \right] \right) && \text{(2a)} \\
\text{s.t.} \quad & \sum_{i \in I} R_{it}(\xi) - \alpha_t = \beta_t^{+}(\xi) - \beta_t^{-}(\xi) \quad \forall t \in T && \text{(2b)} \\
& \sum_{i \in I} R_{it}(\xi) \geqslant \beta_t^{+}(\xi) \quad \forall t \in T && \text{(2c)} \\
& \beta_t^{+}(\xi) \leqslant M z_t(\xi), \quad \beta_t^{-}(\xi) \leqslant M(1 - z_t(\xi)) \quad \forall t \in T && \text{(2d)} \\
& \alpha_t^{DA} \geqslant 0, \beta_t^{+}(\xi) \geqslant 0, \beta_t^{-}(\xi) \geqslant 0, z_t(\xi) \in \{0, 1\} \quad \forall t \in T && \text{(2e)}
\end{aligned}
$$

1.3 Settlement

$$
\begin{aligned}
\max \quad & \sum_{t \in T} \left(P_t^{DA} \alpha_t + \mathbb{E}\left[P_t^{RT}(\xi) \beta_t^{+}(\xi) - P_t^{PN} \beta_t^{-}(\xi) \right] \right) && \text{(3a)} \\
\text{s.t.} \quad & \sum_{i \in I} R_{it}(\xi) - \alpha_t^{DA} = \beta_t^{+}(\xi) - \beta_t^{-}(\xi) \quad \forall t \in T && \text{(3b)} \\
& \sum_{i \in I} R_{it}(\xi) \geqslant \beta_t^{+}(\xi) \quad \forall t \in T && \text{(3c)} \\
& \beta_t^{+}(\xi) \leqslant M z_t(\xi), \quad \beta_t^{-}(\xi) \leqslant M(1 - z_t(\xi)) \quad \forall t \in T && \text{(3d)} \\
& \alpha_t = \sum_{i \in I} x_{it}(\xi), \quad \beta_t^{+}(\xi) = \sum_{i \in I} e_{it}^{+}(\xi), \quad \beta_t^{-}(\xi) = \sum_{i \in I} e_{it}^{-}(\xi) \quad \forall t \in T && \text{(3e)} \\
& R_{it}(\xi) - x_{it}(\xi) = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) \quad \forall t \in T && \text{(3f)} \\
& R_{it}(\xi) \geqslant y_{it}^{+}(\xi) \quad \forall t \in T && \text{(3g)} \\
& y_{it}^{+}(\xi) \leqslant M z_{it}(\xi), \quad y_{it}^{-}(\xi) \leqslant M(1 - z_{it}(\xi)) \quad \forall t \in T && \text{(3h)} \\
& \sum_{j \in I, j \neq i} d_{ijt}(\xi) \leqslant y_{it}^{+}(\xi), \quad \sum_{j \in I, j \neq i} d_{jit}(\xi) \leqslant y_{it}^{-}(\xi) \quad \forall t \in T && \text{(3i)} \\
& d_{iit}(\xi) = 0 \quad \forall t \in T && \text{(3j)} \\
& e_{it}^{+}(\xi) = y_{it}^{+}(\xi) - \sum_{j \in I, j \neq i} d_{ijt}(\xi) \quad \forall t \in T && \text{(3k)} \\
& e_{it}^{-}(\xi) = y_{it}^{-}(\xi) - \sum_{j \in I, j \neq i} d_{jit}(\xi) \quad \forall t \in T && \text{(3l)}
\end{aligned}
$$

2 With Storage

2.1 Disaggregation

$$
\begin{aligned}
\text{max} \quad & \sum_{t \in T} \left(P_t^{DA} x_{it} + \mathbb{E} \left[P_t^{RT}(\xi) y_{it}^{+}(\xi) - P_t^{PN} y_{it}^{-}(\xi) \right] \right) && \text{(4a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \text{(4b)} \\
& R_{it}(\xi) \geqslant y_{it}^{+}(\xi) && \text{(4c)} \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) \quad \forall i, t && \text{(4d)} \\
& z_{it}^{D}(\xi) \leqslant z_{it}(\xi), \quad z_{it}^{C}(\xi) \leqslant K_i - z_{it}(\xi), \quad 0 \leqslant z_{it}(\xi) \leqslant K_i && \text{(4e)} \\
& y_{it}^{+}(\xi) \leqslant M_1 \phi_{it}^{1}(\xi), \quad y_{it}^{-}(\xi) \leqslant M_1 (1 - \phi_{it}^{1}(\xi)) && \text{(4f)} \\
& y_{it}^{-}(\xi) \leqslant M_1 \phi_{it}^{2}(\xi), \quad z_{it}^{C}(\xi) \leqslant M_1 (1 - \phi_{it}^{2}(\xi)) && \text{(4g)} \\
& z_{it}^{C}(\xi) \leqslant M_1 \phi_{it}^{3}(\xi), \quad z_{it}^{D}(\xi) \leqslant M_1 (1 - \phi_{it}^{3}(\xi)) && \text{(4h)}
\end{aligned}
$$

2.2 Aggregation with individual BTM storage control - y, d, e correlated

$$
\begin{aligned}
\text{max} \quad & \sum_{t \in T} \left(P_t^{DA} \sum_{i \in I} x_{it} + \mathbb{E} \left[P_t^{RT}(\xi) \sum_{i \in I} e_{it}^{+}(\xi) - P_t^{PN} \sum_{i \in I} e_{it}^{-}(\xi) \right] \right) && \text{(5a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \text{(5b)} \\
& R_{it}(\xi) + z_{it}^{D}(\xi) \geqslant y_{it}^{+}(\xi) + z_{it}^{C}(\xi) && \text{(5c)} \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + \eta^{C} z_{it}^{C}(\xi) - \frac{1}{\eta^{D}} z_{it}^{D}(\xi) && \text{(5d)} \\
& z_{it}^{D}(\xi) \leqslant z_{it}(\xi), \quad z_{it}^{C}(\xi) \leqslant K_i - z_{it}(\xi), \quad 0 \leqslant z_{it}(\xi) \leqslant K_i && \text{(5e)} \\
& e_{it}^{+}(\xi) = y_{it}^{+}(\xi) - d_{it}^{+}(\xi), \quad e_{it}^{-}(\xi) = y_{it}^{-}(\xi) - d_{it}^{-}(\xi) && \text{(5f)} \\
& \sum_{i \in I} d_{it}^{+}(\xi) = \sum_{i \in I} d_{it}^{-}(\xi) && \text{(5g)} \\
& y_{it}^{+}(\xi) \leqslant M_1 \phi_{it}^{1}(\xi), \quad y_{it}^{-}(\xi) \leqslant M_1 (1 - \phi_{it}^{1}(\xi)) && \text{(5h)} \\
& y_{it}^{-}(\xi) \leqslant M_1 \phi_{it}^{2}(\xi), \quad z_{it}^{C}(\xi) \leqslant M_1 (1 - \phi_{it}^{2}(\xi)) && \text{(5i)} \\
& z_{it}^{C}(\xi) \leqslant M_1 \phi_{it}^{3}(\xi), \quad z_{it}^{D}(\xi) \leqslant M_1 (1 - \phi_{it}^{3}(\xi)) && \text{(5j)}
\end{aligned}
$$

2.3 Holistic Aggregation with individual BTM storage control - y, d independent

$$
\begin{aligned}
\text{max} \quad & \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \mathbb{E} \left[P_t^{RT}(\xi) \cdot y_{it}^{+}(\xi) - P_t^{PN}(\xi) \cdot y_{it}^{-}(\xi) \right] \right) && && \text{(6a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) + d_{it}^{+}(\xi) - d_{it}^{-}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \forall i \in I, t \in T && \text{(6b)} \\
& R_{it}(\xi) + z_{it}^{D}(\xi) \geqslant y_{it}^{+}(\xi) + d_{it}^{+}(\xi) + z_{it}^{C}(\xi) && \forall i \in I, t \in T && \text{(6c)} \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + \eta^{C} z_{it}^{C}(\xi) - \frac{1}{\eta^{D}} z_{it}^{D}(\xi) && \forall i \in I, t \in T && \text{(6d)} \\
& z_{it}^{D}(\xi) \leqslant z_{it}(\xi), \quad z_{it}^{C}(\xi) \leqslant K_i - z_{it}(\xi), \quad 0 \leqslant z_{it}(\xi) \leqslant K_i && \forall i \in I, t \in T && \text{(6e)} \\
& y_{it}^{+}(\xi) \leqslant M \phi_{it}^{1}(\xi), \quad y_{it}^{-}(\xi) \leqslant M (1 - \phi_{it}^{1}(\xi)) && \forall i \in I, t \in T && \text{(6f)} \\
& y_{it}^{-}(\xi) \leqslant M \phi_{it}^{2}(\xi), \quad z_{it}^{C}(\xi) \leqslant M (1 - \phi_{it}^{2}(\xi)) && \forall i \in I, t \in T && \text{(6g)} \\
& z_{it}^{C}(\xi) \leqslant M \phi_{it}^{3}(\xi), \quad z_{it}^{D}(\xi) \leqslant M (1 - \phi_{it}^{3}(\xi)) && \forall i \in I, t \in T && \text{(6h)} \\
& d_{it}^{+}(\xi) \leqslant M \phi_{it}^{4}(\xi), \quad d_{it}^{-}(\xi) \leqslant M (1 - \phi_{it}^{4}(\xi)) && \forall i \in I, t \in T && \text{(6i)} \\
& z_{it}^{C}(\xi) \leqslant M \phi_{it}^{5}(\xi), \quad d_{it}^{-}(\xi) \leqslant M (1 - \phi_{it}^{5}(\xi)) && \forall i \in I, t \in T && \text{(6j)} \\
& y_{it}^{+}(\xi) \leqslant M \phi_{it}^{6}(\xi), \quad d_{it}^{-}(\xi) \leqslant M (1 - \phi_{it}^{6}(\xi)) && \forall i \in I, t \in T && \text{(6k)} \\
& y_{it}^{-}(\xi) \leqslant M \phi_{it}^{7}(\xi), \quad d_{it}^{+}(\xi) \leqslant M (1 - \phi_{it}^{7}(\xi)) && \forall i \in I, t \in T && \text{(6l)} \\
& \sum_{i \in I} d_{it}^{+}(\xi) = \sum_{i \in I} d_{it}^{-}(\xi) \quad : \quad \lambda_t(\xi) && \forall t \in T && \text{(6m)} \\
& \Pi_i^{IND} \leqslant \Pi_i^{HOL} && \forall i \in I && \text{(6n)}
\end{aligned}
$$

2.4 Aggregation with direct control over storage

$$\max \quad \sum_{t \in T} \left(P_t^{DA} \alpha_t + \mathbb{E}\left[P_t^{RT}(\xi) \beta_t^+(\xi) - P_t^{PN} \beta_t^-(\xi)) \right] \right) \tag{7a}$$

$$\text{s.t.} \quad \sum_{i \in I} R_{it}(\xi) - \alpha_t = \beta_t^+(\xi) - \beta_t^-(\xi) + \gamma_t^C(\xi) - \gamma_t^D(\xi) \quad \forall t \tag{7b}$$

$$\sum_{i \in I} R_{it}(\xi) \geqslant \beta_t^+(\xi) \tag{7c}$$

$$\gamma_t^D(\xi) \leqslant \gamma_t(\xi), \quad \gamma_t^C(\xi) \leqslant \sum_{i \in I} K_i - \gamma_t(\xi), \quad 0 \leqslant \gamma_t(\xi) \leqslant \sum_{i \in I} K_i \quad \forall t \tag{7d}$$

$$\gamma_{t+1}(\xi) = \gamma_t(\xi) + \gamma_t^C(\xi) - \gamma_t^D(\xi) \quad \forall t \tag{7e}$$

$$\beta_t^+(\xi) \leqslant M_2 \mu_t(\xi), \quad \beta_t^-(\xi) \leqslant M_2(1 - \mu_t(\xi)) \quad \forall t \tag{7f}$$

$$\beta_t^-(\xi) \leqslant M_2 \eta_t(\xi), \quad \gamma_t^C(\xi) \leqslant M_2(1 - \eta_t(\xi)) \quad \forall t \tag{7g}$$

$$\gamma_t^C(\xi) \leqslant M_2 \lambda_t(\xi), \quad \gamma_t^D(\xi) \leqslant M_2(1 - \lambda_t(\xi)) \quad \forall t \tag{7h}$$

3 Individual

3.1 Different Internal Price (Non-linear)

$$\max \quad \sum_{t\in T} \left(P_t^{DA} \cdot x_t + \mathbb{E}\left[P_t^{RT}(\xi) \cdot y_t^+(\xi) - P_t^{PN} \cdot y_t^-(\xi) + \rho_t^+(d) \cdot d_t^+(\xi) - \rho_t^-(d) \cdot d_t^-(\xi)\right]\right) \tag{8a}$$

$$\text{s.t.} \quad R_t(\xi) - x_t = y_t^+(\xi) - y_t^-(\xi) + d_t^+(\xi) - d_t^-(\xi) + z_t^C(\xi) - z_t^D(\xi) \tag{8b}$$

$$R_t(\xi) + z_t^D(\xi) \geqslant y_t^+(\xi) + d_t^+(\xi) + z_t^C(\xi) \tag{8c}$$

$$z_{t+1}(\xi) = z_t(\xi) + z_t^C(\xi) - z_t^D(\xi) \tag{8d}$$

$$z_t^D(\xi) \leqslant z_t(\xi), \quad z_t^C(\xi) \leqslant K - z_t(\xi), \quad 0 \leqslant z_t(\xi) \leqslant K \tag{8e}$$

$$y_t^+(\xi) \leqslant M\phi_t^1(\xi), \quad y_t^-(\xi) \leqslant M(1 - \phi_t^1(\xi)) \tag{8f}$$

$$y_t^-(\xi) \leqslant M\phi_t^2(\xi), \quad z_t^C(\xi) \leqslant M(1 - \phi_t^2(\xi)) \tag{8g}$$

$$z_t^C(\xi) \leqslant M\phi_t^3(\xi), \quad z_t^D(\xi) \leqslant M(1 - \phi_t^3(\xi)) \tag{8h}$$

$$d_t^+(\xi) \leqslant M\phi_t^4(\xi), \quad d_t^-(\xi) \leqslant M(1 - \phi_t^4(\xi)) \tag{8i}$$

3.2 Different Internal Price (Stepwise-Linear)

$$\begin{aligned}\max \quad \sum_{t\in T} \Bigg(& P_t^{DA} \cdot x_t + \mathbb{E}\Big[P_t^{RT}(\xi) \cdot y_t^+(\xi) - P_t^{PN} \cdot y_t^-(\xi) \\ & + \sum_{b^+\in B_t^+} \rho_{t,b^+}^+ \cdot (w_{t,b^+}^+(\xi) + u_{t,b^+}^+(\xi) \cdot D_{t,b^+}^{\text{min}\,+}) - \sum_{b^-\in B_t^-} \rho_{t,b^-}^- \cdot (w_{t,b^-}^-(\xi) + u_{t,b^-}^-(\xi) \cdot D_{t,b^-}^{\text{min}\,-})\Big]\Bigg)\end{aligned} \tag{9a}$$

$$\text{s.t.} \quad d_t^+(\xi) = \sum_{b^+\in B_t^+} (w_{t,b^+}^+(\xi) + u_{t,b^+}^+(\xi) \cdot D_{t,b^+}^{\text{min}\,+}) \qquad \forall t \tag{9b}$$

$$0 \leqslant w_{t,b^+}^+(\xi) \leqslant u_{t,b^+}^+(\xi) \cdot W_{t,b^+}^{\text{max}\,+} \qquad \forall t, b^+ \tag{9c}$$

$$\sum_{b^+\in B_t^+} u_{t,b^+}^+(\xi) \leqslant 1 \qquad \forall t \tag{9d}$$

$$d_t^-(\xi) = \sum_{b^-\in B_t^-} (w_{t,b^-}^-(\xi) + u_{t,b^-}^-(\xi) \cdot D_{t,b^-}^{\text{min}\,-}) \qquad \forall t \tag{9e}$$

$$0 \leqslant w_{t,b^-}^-(\xi) \leqslant u_{t,b^-}^-(\xi) \cdot W_{t,b^-}^{\text{max}\,-} \qquad \forall t, b^- \tag{9f}$$

$$\sum_{b^-\in B_t^-} u_{t,b^-}^-(\xi) \leqslant 1 \qquad \forall t \tag{9g}$$

$$R_t(\xi) - x_t = y_t^+(\xi) - y_t^-(\xi) + d_t^+(\xi) - d_t^-(\xi) + z_t^C(\xi) - z_t^D(\xi) \qquad \forall t \tag{9h}$$

$$R_t(\xi) + z_t^D(\xi) \geqslant y_t^+(\xi) + d_t^+(\xi) + z_t^C(\xi) \qquad \forall t \tag{9i}$$

$$z_{t+1}(\xi) = z_t(\xi) + z_t^C(\xi) - z_t^D(\xi) \qquad \forall t \tag{9j}$$

$$z_t^D(\xi) \leqslant z_t(\xi), \quad z_t^C(\xi) \leqslant K - z_t(\xi), \quad 0 \leqslant z_t(\xi) \leqslant K \qquad \forall t \tag{9k}$$

$$y_t^+(\xi) \leqslant M\phi_t^1(\xi), \quad y_t^-(\xi) \leqslant M(1 - \phi_t^1(\xi)) \qquad \forall t \tag{9l}$$

$$y_t^-(\xi) \leqslant M\phi_t^2(\xi), \quad z_t^C(\xi) \leqslant M(1 - \phi_t^2(\xi)) \qquad \forall t \tag{9m}$$

$$z_t^C(\xi) \leqslant M\phi_t^3(\xi), \quad z_t^D(\xi) \leqslant M(1 - \phi_t^3(\xi)) \qquad \forall t \tag{9n}$$

$$d_t^+(\xi) \leqslant M\phi_t^4(\xi), \quad d_t^-(\xi) \leqslant M(1 - \phi_t^4(\xi)) \qquad \forall t \tag{9o}$$

4 Endogeneous Pricing

4.1 Price as DV

$$
\begin{aligned}
\max \quad & \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \mathbb{E}\left[P_t^{RT}(\xi) \cdot y_{it}^{+}(\xi) - P_t^{PN}(\xi) \cdot y_{it}^{-}(\xi) + \rho_t^{+} \cdot d_{it}^{+}(\xi) - \rho_t^{-} \cdot d_{it}^{-}(\xi) \right] \right) && && \text{(10a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) + d_{it}^{+}(\xi) - d_{it}^{-}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \forall i \in I, t \in T && \text{(10b)} \\
& R_{it}(\xi) + z_{it}^{D}(\xi) \geqslant y_{it}^{+}(\xi) + d_{it}^{+}(\xi) + z_{it}^{C}(\xi) && \forall i \in I, t \in T && \text{(10c)} \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \forall i \in I, t \in T && \text{(10d)} \\
& z_{it}^{D}(\xi) \leqslant z_{it}(\xi), \quad z_{it}^{C}(\xi) \leqslant K_i - z_{it}(\xi), \quad 0 \leqslant z_{it}(\xi) \leqslant K_i && \forall i \in I, t \in T && \text{(10e)} \\
& y_{it}^{+}(\xi) \leqslant M\phi_{it}^{1}(\xi), \quad y_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{1}(\xi)) && \forall i \in I, t \in T && \text{(10f)} \\
& y_{it}^{-}(\xi) \leqslant M\phi_{it}^{2}(\xi), \quad z_{it}^{C}(\xi) \leqslant M(1 - \phi_{it}^{2}(\xi)) && \forall i \in I, t \in T && \text{(10g)} \\
& z_{it}^{C}(\xi) \leqslant M\phi_{it}^{3}(\xi), \quad z_{it}^{D}(\xi) \leqslant M(1 - \phi_{it}^{3}(\xi)) && \forall i \in I, t \in T && \text{(10h)} \\
& d_{it}^{+}(\xi) \leqslant M\phi_{it}^{4}(\xi), \quad d_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{4}(\xi)) && \forall i \in I, t \in T && \text{(10i)} \\
& z_{it}^{C}(\xi) \leqslant M\phi_{it}^{5}(\xi), \quad d_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{5}(\xi)) && \forall i \in I, t \in T && \text{(10j)} \\
& y_{it}^{+}(\xi) \leqslant M\phi_{it}^{6}(\xi), \quad d_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{6}(\xi)) && \forall i \in I, t \in T && \text{(10k)} \\
& y_{it}^{-}(\xi) \leqslant M\phi_{it}^{7}(\xi), \quad d_{it}^{+}(\xi) \leqslant M(1 - \phi_{it}^{7}(\xi)) && \forall i \in I, t \in T && \text{(10l)} \\
& \sum_{i \in I} d_{it}^{+}(\xi) = \sum_{i \in I} d_{it}^{-}(\xi) && \forall t \in T && \text{(10m)} \\
& \rho_t^{MIN} \leqslant \rho_t^{+}, \quad \rho_t^{-} \leqslant \rho_t^{MAX} && \forall t \in T && \text{(10n)} \\
& \Pi_i^{END} = \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \mathbb{E}\left[P_t^{RT}(\xi) \cdot y_{it}^{+}(\xi) - P_t^{PN}(\xi) \cdot y_{it}^{-}(\xi) + \rho_t^{+} \cdot d_{it}^{+}(\xi) - \rho_t^{-} \cdot d_{it}^{-}(\xi) \right] \right) && \forall i \in I && \text{(10o)} \\
& \Pi_i^{IND} \leqslant \Pi_i^{END} && \forall i \in I && \text{(10p)} \\
& \sum_{i \in I} \Pi_i^{END} \leqslant \Pi^{HOL} && && \text{(10q)}
\end{aligned}
$$

4.2 Price as DV - Market Clearing ver.

$$
\begin{aligned}
\max \quad & \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \mathbb{E}\left[P_t^{RT}(\xi) \cdot y_{it}^{+}(\xi) - P_t^{PN}(\xi) \cdot y_{it}^{-}(\xi) + \rho_t \cdot d_{it}^{+}(\xi) - \rho_t \cdot d_{it}^{-}(\xi) \right] \right) && && \text{(11a)} \\
\text{s.t.} \quad & R_{it}(\xi) - x_{it} = y_{it}^{+}(\xi) - y_{it}^{-}(\xi) + d_{it}^{+}(\xi) - d_{it}^{-}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \forall i \in I, t \in T && \text{(11b)} \\
& R_{it}(\xi) + z_{it}^{D}(\xi) \geqslant y_{it}^{+}(\xi) + d_{it}^{+}(\xi) + z_{it}^{C}(\xi) && \forall i \in I, t \in T && \text{(11c)} \\
& z_{i,t+1}(\xi) = z_{it}(\xi) + z_{it}^{C}(\xi) - z_{it}^{D}(\xi) && \forall i \in I, t \in T && \text{(11d)} \\
& z_{it}^{D}(\xi) \leqslant z_{it}(\xi), \quad z_{it}^{C}(\xi) \leqslant K_i - z_{it}(\xi), \quad 0 \leqslant z_{it}(\xi) \leqslant K_i && \forall i \in I, t \in T && \text{(11e)} \\
& y_{it}^{+}(\xi) \leqslant M\phi_{it}^{1}(\xi), \quad y_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{1}(\xi)) && \forall i \in I, t \in T && \text{(11f)} \\
& y_{it}^{-}(\xi) \leqslant M\phi_{it}^{2}(\xi), \quad z_{it}^{C}(\xi) \leqslant M(1 - \phi_{it}^{2}(\xi)) && \forall i \in I, t \in T && \text{(11g)} \\
& z_{it}^{C}(\xi) \leqslant M\phi_{it}^{3}(\xi), \quad z_{it}^{D}(\xi) \leqslant M(1 - \phi_{it}^{3}(\xi)) && \forall i \in I, t \in T && \text{(11h)} \\
& d_{it}^{+}(\xi) \leqslant M\phi_{it}^{4}(\xi), \quad d_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{4}(\xi)) && \forall i \in I, t \in T && \text{(11i)} \\
& z_{it}^{C}(\xi) \leqslant M\phi_{it}^{5}(\xi), \quad d_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{5}(\xi)) && \forall i \in I, t \in T && \text{(11j)} \\
& y_{it}^{+}(\xi) \leqslant M\phi_{it}^{6}(\xi), \quad d_{it}^{-}(\xi) \leqslant M(1 - \phi_{it}^{6}(\xi)) && \forall i \in I, t \in T && \text{(11k)} \\
& y_{it}^{-}(\xi) \leqslant M\phi_{it}^{7}(\xi), \quad d_{it}^{+}(\xi) \leqslant M(1 - \phi_{it}^{7}(\xi)) && \forall i \in I, t \in T && \text{(11l)} \\
& \sum_{i \in I} d_{it}^{+}(\xi) = \sum_{i \in I} d_{it}^{-}(\xi) && \forall t \in T && \text{(11m)} \\
& \rho_t^{MIN} \leqslant \rho_t \leqslant \rho_t^{MAX} && \forall t \in T && \text{(11n)} \\
& \Pi_i^{END} = \sum_{t \in T} \left(P_t^{DA} \cdot x_{it} + \mathbb{E}\left[P_t^{RT}(\xi) \cdot y_{it}^{+}(\xi) - P_t^{PN}(\xi) \cdot y_{it}^{-}(\xi) + \rho_t^{+} \cdot d_{it}^{+}(\xi) - \rho_t^{-} \cdot d_{it}^{-}(\xi) \right] \right) && \forall i \in I && \text{(11o)} \\
& \Pi_i^{IND} \leqslant \Pi_i^{END} && \forall i \in I && \text{(11p)}
\end{aligned}
$$

5 Lagrangian Dual

5.1 Holistic Aggregation

$$\begin{aligned}
\max \quad & \sum_{i\in I}\sum_{t\in T}\left(P_t^{DA}\cdot x_{it}+\frac{1}{S}\sum_{s\in S}\left(P_t^{RT}(\xi_s)\cdot y_{it}^{+}(\xi_s)-P_t^{PN}(\xi_s)\cdot y_{it}^{-}(\xi_s)\right)\right) && && (12a)\\
\text{s.t.} \quad & R_{it}(\xi_s)-x_{it}=y_{it}^{+}(\xi_s)-y_{it}^{-}(\xi_s)+d_{it}^{+}(\xi_s)-d_{it}^{-}(\xi_s)+z_{it}^{C}(\xi_s)-z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && (12b)\\
& R_{it}(\xi_s)+z_{it}^{D}(\xi_s)\geqslant y_{it}^{+}(\xi_s)+d_{it}^{+}(\xi_s)+z_{it}^{C}(\xi_s) && \forall i\in I,t\in T,s\in S && (12c)\\
& z_{i,t+1}(\xi_s)=z_{it}(\xi_s)+\eta^{C}z_{it}^{C}(\xi_s)-\frac{1}{\eta^{D}}z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && (12d)\\
& z_{it}^{D}(\xi_s)\leqslant z_{it}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant K_i-z_{it}(\xi_s),\quad 0\leqslant z_{it}(\xi_s)\leqslant K_i && \forall i\in I,t\in T,s\in S && (12e)\\
& \sum_{i\in I}d_{it}^{+}(\xi_s)=\sum_{i\in I}d_{it}^{-}(\xi_s)\quad :\quad \lambda_t(\xi_s) && \forall t\in T,s\in S && (12f)\\
& y_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{1}(\xi_s),\quad y_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{1}(\xi_s)) && \forall i\in I,t\in T,s\in S && (12g)\\
& d_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{2}(\xi_s),\quad d_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{2}(\xi_s)) && \forall i\in I,t\in T,s\in S && (12h)\\
& y_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{3}(\xi_s),\quad d_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{3}(\xi_s)) && \forall i\in I,t\in T,s\in S && (12i)\\
& y_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{4}(\xi_s),\quad d_{it}^{+}(\xi_s)\leqslant M(1-\phi_{it}^{4}(\xi_s)) && \forall i\in I,t\in T,s\in S && (12j)\\
& y_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{5}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant M(1-\phi_{it}^{5}(\xi_s)) && \forall i\in I,t\in T,s\in S && (12k)\\
& d_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{6}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant M(1-\phi_{it}^{6}(\xi_s)) && \forall i\in I,t\in T,s\in S && (12l)
\end{aligned}$$

(**??**): Maximizing total expected revenue of the aggregation from day-ahead trading and real-time balancing.
(**??**): Energy balance equation ensuring the net energy flows regarding day-ahead commitment, real-time trading, internal trading, and battery operations equals the amount of generation.
(**??**): Feasibility constraint ensuring that total energy outflows do not exceed available energy.
(**??**): Battery state-of-charge dynamics with charging and discharge efficiency.
(**??**): Battery operational limits ensuring operations do not exceed currently available capacity.
(**??**): Internal trading balance constraint ensuring internal pool supply and demand are cleared, with dual variable $\lambda_t(\xi_s)$ representing the internal settlement price.
(**??**)-(**??**): Arbitrage control measures preventing the simultaneous buying and selling of energy within and among markets
(**??**)-(**??**): Arbitrage control measures prohibiting the purchase of energy for the purpose of charging the battery

5.2 Lagrangian Relaxed Holistic Aggregation

$$\begin{aligned}
\max \quad & \sum_{i\in I}\sum_{t\in T}\left(P_t^{DA}\cdot x_{it}+\frac{1}{S}\sum_{s\in S}\left(P_t^{RT}(\xi_s)\cdot y_{it}^{+}(\xi_s)-P_t^{PN}(\xi_s)\cdot y_{it}^{-}(\xi_s)\right)\right) && && (13a)\\
& +\sum_{s\in S}\sum_{t\in T}\lambda_t(\xi_s)\left(\sum_{i\in I}d_{it}^{-}(\xi_s)-\sum_{i\in I}d_{it}^{+}(\xi_s)\right) && && \\
\text{s.t.} \quad & R_{it}(\xi_s)-x_{it}=y_{it}^{+}(\xi_s)-y_{it}^{-}(\xi_s)+d_{it}^{+}(\xi_s)-d_{it}^{-}(\xi_s)+z_{it}^{C}(\xi_s)-z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && (13b)\\
& R_{it}(\xi_s)+z_{it}^{D}(\xi_s)\geqslant y_{it}^{+}(\xi_s)+d_{it}^{+}(\xi_s)+z_{it}^{C}(\xi_s) && \forall i\in I,t\in T,s\in S && (13c)\\
& z_{i,t+1}(\xi_s)=z_{it}(\xi_s)+\eta^{C}z_{it}^{C}(\xi_s)-\frac{1}{\eta^{D}}z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && (13d)\\
& z_{it}^{D}(\xi_s)\leqslant z_{it}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant K_i-z_{it}(\xi_s),\quad 0\leqslant z_{it}(\xi_s)\leqslant K_i && \forall i\in I,t\in T,s\in S && (13e)\\
& y_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{1}(\xi_s),\quad y_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{1}(\xi_s)) && \forall i\in I,t\in T,s\in S && (13f)\\
& d_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{2}(\xi_s),\quad d_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{2}(\xi_s)) && \forall i\in I,t\in T,s\in S && (13g)\\
& y_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{3}(\xi_s),\quad d_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{3}(\xi_s)) && \forall i\in I,t\in T,s\in S && (13h)\\
& y_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{4}(\xi_s),\quad d_{it}^{+}(\xi_s)\leqslant M(1-\phi_{it}^{4}(\xi_s)) && \forall i\in I,t\in T,s\in S && (13i)\\
& y_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{5}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant M(1-\phi_{it}^{5}(\xi_s)) && \forall i\in I,t\in T,s\in S && (13j)\\
& d_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{6}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant M(1-\phi_{it}^{6}(\xi_s)) && \forall i\in I,t\in T,s\in S && (13k)
\end{aligned}$$

5.3 Individual Replay

for each i:

$$
\begin{aligned}
\max \quad & \sum_{t\in T}\left(P_t^{DA}x_t + \frac{1}{S}\sum_{s\in S}\left(P_t^{RT}(\xi_s)y_t^+(\xi_s) - P_t^{PN}(\xi_s)y_t^-(\xi_s)\right)\right) + \sum_{s\in S}\sum_{t\in T}\lambda_t(\xi_s)\left(d_t^-(\xi_s) - d_t^+(\xi_s)\right) && && \text{(14a)}\\
\text{s.t.} \quad & R_t(\xi_s) - x_t = y_t^+(\xi_s) - y_t^-(\xi_s) + d_t^+(\xi_s) - d_t^-(\xi_s) + z_t^C(\xi_s) - z_t^D(\xi_s) && \forall t\in T, s\in S && \text{(14b)}\\
& R_t(\xi_s) + z_t^D(\xi_s) \geqslant y_t^+(\xi_s) + d_t^+(\xi_s) + z_t^C(\xi_s) && \forall t\in T, s\in S && \text{(14c)}\\
& z_{t+1}(\xi_s) = z_t(\xi_s) + \eta^C z_t^C(\xi_s) - \frac{1}{\eta^D}z_t^D(\xi_s) && \forall t\in T, s\in S && \text{(14d)}\\
& z_t^D(\xi_s) \leqslant z_t(\xi_s), \quad z_t^C(\xi_s) \leqslant K - z_t(\xi_s), \quad 0 \leqslant z_t(\xi_s) \leqslant K && \forall t\in T, s\in S && \text{(14e)}\\
& y_t^+(\xi_s) \leqslant M\phi_t^1(\xi_s), \quad y_t^-(\xi_s) \leqslant M(1-\phi_t^1(\xi_s)) && \forall t\in T, s\in S && \text{(14f)}\\
& d_t^+(\xi_s) \leqslant M\phi_t^2(\xi_s), \quad d_t^-(\xi_s) \leqslant M(1-\phi_t^2(\xi_s)) && \forall t\in T, s\in S && \text{(14g)}\\
& y_t^+(\xi_s) \leqslant M\phi_t^3(\xi_s), \quad d_t^-(\xi_s) \leqslant M(1-\phi_t^3(\xi_s)) && \forall t\in T, s\in S && \text{(14h)}\\
& y_t^-(\xi_s) \leqslant M\phi_t^4(\xi_s), \quad d_t^+(\xi_s) \leqslant M(1-\phi_t^4(\xi_s)) && \forall t\in T, s\in S && \text{(14i)}\\
& y_t^-(\xi_s) \leqslant M\phi_t^5(\xi_s), \quad z_t^C(\xi_s) \leqslant M(1-\phi_t^5(\xi_s)) && \forall t\in T, s\in S && \text{(14j)}\\
& d_t^-(\xi_s) \leqslant M\phi_t^6(\xi_s), \quad z_t^C(\xi_s) \leqslant M(1-\phi_t^6(\xi_s)) && \forall t\in T, s\in S && \text{(14k)}
\end{aligned}
$$

6 Linear Decision Rule

6.1 No Arbitrage Control

$$
\begin{aligned}
\max \quad & \sum_{i\in I}\sum_{t\in T}\left(P_t^{DA}\cdot x_{it}+\frac{1}{S}\sum_{s\in S}\left(P_t^{RT}(\xi_s)\cdot y_{it}^{+}(\xi_s)-P_t^{PN}(\xi_s)\cdot y_{it}^{-}(\xi_s)\right)\right) && && \text{(15a)}\\
\text{s.t.} \quad & R_{it}(\xi_s)-x_{it}=y_{it}^{+}(\xi_s)-y_{it}^{-}(\xi_s)+d_{it}^{+}(\xi_s)-d_{it}^{-}(\xi_s)+z_{it}^{C}(\xi_s)-z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15b)}\\
& R_{it}(\xi_s)+z_{it}^{D}(\xi_s)\geqslant y_{it}^{+}(\xi_s)+d_{it}^{+}(\xi_s)+z_{it}^{C}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15c)}\\
& z_{i,t+1}(\xi_s)=z_{it}(\xi_s)+\eta^{C}z_{it}^{C}(\xi_s)-\frac{1}{\eta^{D}}z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15d)}\\
& z_{it}^{D}(\xi_s)\leqslant z_{it}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant K_i-z_{it}(\xi_s),\quad 0\leqslant z_{it}(\xi_s)\leqslant K_i && \forall i\in I,t\in T,s\in S && \text{(15e)}\\
& \sum_{i\in I}d_{it}^{+}(\xi_s)=\sum_{i\in I}d_{it}^{-}(\xi_s)\quad:\quad \lambda_t(\xi_s) && \forall t\in T,s\in S && \text{(15f)}\\
& y_{it}^{+}(\xi_s)=y_{it,0}^{+}+y_{it,R}^{+}\cdot R_{it}(\xi_s)+y_{it,RT}^{+}\cdot P_t^{RT}(\xi_s)+y_{it,PN}^{+}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15g)}\\
& y_{it}^{-}(\xi_s)=y_{it,0}^{-}+y_{it,R}^{-}\cdot R_{it}(\xi_s)+y_{it,RT}^{-}\cdot P_t^{RT}(\xi_s)+y_{it,PN}^{-}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15h)}\\
& d_{it}^{+}(\xi_s)=d_{it,0}^{+}+d_{it,R}^{+}\cdot R_{it}(\xi_s)+d_{it,RT}^{+}\cdot P_t^{RT}(\xi_s)+d_{it,PN}^{+}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15i)}\\
& d_{it}^{-}(\xi_s)=d_{it,0}^{-}+d_{it,R}^{-}\cdot R_{it}(\xi_s)+d_{it,RT}^{-}\cdot P_t^{RT}(\xi_s)+d_{it,PN}^{-}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15j)}\\
& z_{it}^{C}(\xi_s)=z_{it,0}^{C}+z_{it,R}^{C}\cdot R_{it}(\xi_s)+z_{it,RT}^{C}\cdot P_t^{RT}(\xi_s)+z_{it,PN}^{C}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15k)}\\
& z_{it}^{D}(\xi_s)=z_{it,0}^{D}+z_{it,R}^{D}\cdot R_{it}(\xi_s)+z_{it,RT}^{D}\cdot P_t^{RT}(\xi_s)+z_{it,PN}^{D}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(15l)}
\end{aligned}
$$

6.2 With Arbitrage Control

$$
\begin{aligned}
\max \quad & \sum_{i\in I}\sum_{t\in T}\left(P_t^{DA}\cdot x_{it}+\frac{1}{S}\sum_{s\in S}\left(P_t^{RT}(\xi_s)\cdot y_{it}^{+}(\xi_s)-P_t^{PN}(\xi_s)\cdot y_{it}^{-}(\xi_s)\right)\right) && && \text{(16a)}\\
\text{s.t.} \quad & R_{it}(\xi_s)-x_{it}=y_{it}^{+}(\xi_s)-y_{it}^{-}(\xi_s)+d_{it}^{+}(\xi_s)-d_{it}^{-}(\xi_s)+z_{it}^{C}(\xi_s)-z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16b)}\\
& R_{it}(\xi_s)+z_{it}^{D}(\xi_s)\geqslant y_{it}^{+}(\xi_s)+d_{it}^{+}(\xi_s)+z_{it}^{C}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16c)}\\
& z_{i,t+1}(\xi_s)=z_{it}(\xi_s)+\eta^{C}z_{it}^{C}(\xi_s)-\frac{1}{\eta^{D}}z_{it}^{D}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16d)}\\
& z_{it}^{D}(\xi_s)\leqslant z_{it}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant K_i-z_{it}(\xi_s),\quad 0\leqslant z_{it}(\xi_s)\leqslant K_i && \forall i\in I,t\in T,s\in S && \text{(16e)}\\
& \sum_{i\in I}d_{it}^{+}(\xi_s)=\sum_{i\in I}d_{it}^{-}(\xi_s)\quad:\quad \lambda_t(\xi_s) && \forall t\in T,s\in S && \text{(16f)}\\
& y_{it}^{+}(\xi_s)=y_{it,0}^{+}+y_{it,R}^{+}\cdot R_{it}(\xi_s)+y_{it,RT}^{+}\cdot P_t^{RT}(\xi_s)+y_{it,PN}^{+}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16g)}\\
& y_{it}^{-}(\xi_s)=y_{it,0}^{-}+y_{it,R}^{-}\cdot R_{it}(\xi_s)+y_{it,RT}^{-}\cdot P_t^{RT}(\xi_s)+y_{it,PN}^{-}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16h)}\\
& d_{it}^{+}(\xi_s)=d_{it,0}^{+}+d_{it,R}^{+}\cdot R_{it}(\xi_s)+d_{it,RT}^{+}\cdot P_t^{RT}(\xi_s)+d_{it,PN}^{+}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16i)}\\
& d_{it}^{-}(\xi_s)=d_{it,0}^{-}+d_{it,R}^{-}\cdot R_{it}(\xi_s)+d_{it,RT}^{-}\cdot P_t^{RT}(\xi_s)+d_{it,PN}^{-}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16j)}\\
& z_{it}^{C}(\xi_s)=z_{it,0}^{C}+z_{it,R}^{C}\cdot R_{it}(\xi_s)+z_{it,RT}^{C}\cdot P_t^{RT}(\xi_s)+z_{it,PN}^{C}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16k)}\\
& z_{it}^{D}(\xi_s)=z_{it,0}^{D}+z_{it,R}^{D}\cdot R_{it}(\xi_s)+z_{it,RT}^{D}\cdot P_t^{RT}(\xi_s)+z_{it,PN}^{D}\cdot P_t^{PN}(\xi_s) && \forall i\in I,t\in T,s\in S && \text{(16l)}\\
& y_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{1}(\xi_s),\quad y_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{1}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16m)}\\
& d_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{2}(\xi_s),\quad d_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{2}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16n)}\\
& y_{it}^{+}(\xi_s)\leqslant M\phi_{it}^{3}(\xi_s),\quad d_{it}^{-}(\xi_s)\leqslant M(1-\phi_{it}^{3}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16o)}\\
& y_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{4}(\xi_s),\quad d_{it}^{+}(\xi_s)\leqslant M(1-\phi_{it}^{4}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16p)}\\
& y_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{5}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant M(1-\phi_{it}^{5}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16q)}\\
& d_{it}^{-}(\xi_s)\leqslant M\phi_{it}^{6}(\xi_s),\quad z_{it}^{C}(\xi_s)\leqslant M(1-\phi_{it}^{6}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16r)}\\
& z_{it}^{C}(\xi_s)\leqslant M\phi_{it}^{7}(\xi_s),\quad z_{it}^{D}(\xi_s)\leqslant M(1-\phi_{it}^{7}(\xi_s)) && \forall i\in I,t\in T,s\in S && \text{(16s)}
\end{aligned}
$$

7 Final (DEP)

7.1 Nomenclature

- Sets and Indices:
  - I: Set of individual DERs $i \in I = \{1, 2, \ldots, N^I\}$
  - T: Set of elementary time periods (hours) $t \in T = \{1, 2, \ldots, N^T\}$
  - S: Set of scenarios used to represent the evolution of uncertain parameters $s \in S = \{1, 2, \ldots, N^S\}$
  - B: Set of arbitrage control conditions $b \in B = \{1, 2, \ldots, N^B\}$
- Parameters:
  - P_t^{DA}: Day-ahead market price at time t
  - $P_t^{RT}(\xi_s)$: Real-time market price at time t under scenario s
  - $P_t^{PN}(\xi_s)$: Commitment shortage penalty price at time t under scenario s
  - $P_t^{IN}(\xi_s)$: Internal pool market price at time t under scenario s
  - $R_{it}(\xi_s)$: Amount of renewable energy generation by DER i at time t under scenario s
  - K_i: Maximum level of DER i's energy storage
  - κ_i^C, κ_i^D: Charging and discharging rate of DER i's energy storage
  - η_i^C, η_i^D: Charging and discharging inefficiency of DER i's energy storage
  - $M_{it}(\xi_s)$: Maximum energy trading availability of DER i at time t under scenario s
  - $M'_{it}(\xi_s)$: Maximum energy storage utilization of DER i at time t under scenario s
  - ξ_s: Realization of uncertain parameters under scenario s
  - π_s: Probability of occurrence of scenario s
- Decision Variables:
  - x_{it}: Day-ahead commitment quantity of DER i at time t
  - $\alpha_{it}, \beta_{it}, \gamma_{it}$: Linear decision rule coefficients for charging decisions of DER i at time t
  - $\alpha'_{it}, \beta'_{it}, \gamma'_{it}$: Linear decision rule coefficients for discharging decisions of DER i at time t
  - $y_{it}^{+}(\xi_s)$: Real-time market selling quantity of DER i at time t under scenario s
  - $y_{it}^{-}(\xi_s)$: Commitment shortage quantity subject to penalty of DER i at time t under scenario s
  - $z_{it}^{C}(\xi_s)$: Charging quantity of DER i at time t under scenario s
  - $z_{it}^{D}(\xi_s)$: Discharging quantity of DER i at time t under scenario s
  - $z_{it}(\xi_s)$: Energy storage level of DER i at time t under scenario s
  - $d_{it}^{+}(\xi_s)$: Internal pool selling quantity of DER i at time t under scenario s
  - $d_{it}^{-}(\xi_s)$: Internal pool buying quantity of DER i at time t under scenario s
  - $\lambda_t(\xi_s)$: Lagrangian multiplier for internal pool supply-demand clearance at time t under scenario s
  - $\phi_{it}^{b}(\xi_s)$: Binary variable for arbitrage control condition b for DER i at time t under scenario s
  - $\mu_{it}^{b}(\xi_s), \delta_{it}^{b}(\xi_s)$: Auxiliary variables for convex hull relaxation of condition b for DER i at time t under scenario s

7.2 Individual Participation

$$
\begin{aligned}
\max \quad & \sum_{i \in I} \sum_{t \in T} \left(P_t^{DA} x_{it} + \pi_s \left(P_t^{RT}(\xi_s) y_{it}^{+}(\xi_s) - P_t^{PN}(\xi_s) y_{it}^{-}(\xi_s) \right) \right) && && \text{(17a)} \\
\text{s.t.} \quad & R_{it}(\xi_s) - x_{it} = y_{it}^{+}(\xi_s) - y_{it}^{-}(\xi_s) + z_{it}^{C}(\xi_s) - z_{it}^{D}(\xi_s) && \forall i \in I, t \in T, s \in S && \text{(17b)} \\
& R_{it}(\xi_s) + z_{it}^{D}(\xi_s) \geqslant y_{it}^{+}(\xi_s) + z_{it}^{C}(\xi_s) && \forall i \in I, t \in T, s \in S && \text{(17c)} \\
& z_{i,t+1}(\xi_s) = z_{it}(\xi_s) + \eta^C z_{it}^{C}(\xi_s) - \frac{1}{\eta^D} z_{it}^{D}(\xi_s) && \forall i \in I, t \in T, s \in S && \text{(17d)} \\
& z_{it}^{D}(\xi_s) \leqslant \min\{\kappa_i^D, z_{it}(\xi_s)\} && \forall i \in I, t \in T, s \in S && \text{(17e)} \\
& z_{it}^{C}(\xi_s) \leqslant \min\{\kappa_i^C, K_i - z_{it}(\xi_s)\} && \forall i \in I, t \in T, s \in S && \text{(17f)} \\
& 0 \leqslant z_{it}(\xi_s) \leqslant K_i && \forall i \in I, t \in T, s \in S && \text{(17g)} \\
& z_{it}^{C}(\xi_s) = \alpha_{it} + \beta_{it} \cdot R_{it}(\xi_s) + \gamma_{it} \left(P_t^{RT}(\xi_s) - P_t^{DA} \right) && \forall i \in I, t \in T, s \in S && \text{(17h)} \\
& z_{it}^{D}(\xi_s) = \alpha'_{it} + \beta'_{it} \cdot R_{it}(\xi_s) + \gamma'_{it} \left(P_t^{RT}(\xi_s) - P_t^{DA} \right) && \forall i \in I, t \in T, s \in S && \text{(17i)} \\
& z_{it}^{C}(\xi_s) \leqslant M' \phi_{it}^{1}(\xi_s), \quad z_{it}^{D}(\xi_s) \leqslant M'(1 - \phi_{it}^{1}(\xi_s)) && \forall i \in I, t \in T, s \in S && \text{(17j)} \\
& y_{it}^{-}(\xi_s) \leqslant M \phi_{it}^{2}(\xi_s), \quad z_{it}^{C}(\xi_s) \leqslant M(1 - \phi_{it}^{2}(\xi_s)) && \forall i \in I, t \in T, s \in S && \text{(17k)}
\end{aligned}
$$

7.3 Holistic Aggregation

$$
\begin{aligned}
\text{max} \quad & \sum_{i\in I}\sum_{t\in T}\left(P_t^{DA}x_{it} + \pi_s\left(P_t^{RT}(\xi_s)y_{it}^+(\xi_s) - P_t^{PN}(\xi_s)y_{it}^-(\xi_s)\right)\right) && && \text{(18a)}\\
\text{s.t.} \quad & R_{it}(\xi_s) - x_{it} = y_{it}^+(\xi_s) - y_{it}^-(\xi_s) + d_{it}^+(\xi_s) - d_{it}^-(\xi_s) + z_{it}^C(\xi_s) - z_{it}^D(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18b)}\\
& R_{it}(\xi_s) + z_{it}^D(\xi_s) \geqslant y_{it}^+(\xi_s) + d_{it}^+(\xi_s) + z_{it}^C(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18c)}\\
& z_{i,t+1}(\xi_s) = z_{it}(\xi_s) + \eta^C z_{it}^C(\xi_s) - \frac{1}{\eta^D} z_{it}^D(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18d)}\\
& z_{it}^D(\xi_s) \leqslant \min\{\kappa_i^D, z_{it}(\xi_s)\} && \forall i\in I, t\in T, s\in S && \text{(18e)}\\
& z_{it}^C(\xi_s) \leqslant \min\{\kappa_i^C, K_i - z_{it}(\xi_s)\} && \forall i\in I, t\in T, s\in S && \text{(18f)}\\
& 0 \leqslant z_{it}(\xi_s) \leqslant K_i && \forall i\in I, t\in T, s\in S && \text{(18g)}\\
& \sum_{i\in I} d_{it}^+(\xi_s) = \sum_{i\in I} d_{it}^-(\xi_s) \quad : \quad \lambda_t(\xi_s) && \forall t\in T, s\in S && \text{(18h)}\\
& z_{it}^C(\xi_s) = \alpha_{it} + \beta_{it}\cdot R_{it}(\xi_s) + \gamma_{it}\left(P_t^{RT}(\xi_s) - P_t^{DA}\right) && \forall i\in I, t\in T, s\in S && \text{(18i)}\\
& z_{it}^D(\xi_s) = \alpha_{it}' + \beta_{it}'\cdot R_{it}(\xi_s) + \gamma_{it}'\left(P_t^{RT}(\xi_s) - P_t^{DA}\right) && \forall i\in I, t\in T, s\in S && \text{(18j)}\\
& \mu_{it}^1(\xi_s) + \delta_{it}^1(\xi_s) \leqslant 1, \quad z_{it}^C(\xi_s) = M'\mu_{it}^1(\xi_s), \quad z_{it}^D(\xi_s) = M'\delta_{it}^1(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18k)}\\
& \mu_{it}^2(\xi_s) + \delta_{it}^2(\xi_s) \leqslant 1, \quad y_{it}^-(\xi_s) = M\mu_{it}^2(\xi_s), \quad z_{it}^C(\xi_s) = M'\delta_{it}^2(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18l)}\\
& \mu_{it}^3(\xi_s) + \delta_{it}^3(\xi_s) \leqslant 1, \quad d_{it}^-(\xi_s) = M\mu_{it}^3(\xi_s), \quad z_{it}^C(\xi_s) = M'\delta_{it}^3(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18m)}\\
& \mu_{it}^4(\xi_s) + \delta_{it}^4(\xi_s) \leqslant 1, \quad d_{it}^+(\xi_s) = M\mu_{it}^4(\xi_s), \quad d_{it}^-(\xi_s) = M\delta_{it}^4(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18n)}\\
& \mu_{it}^5(\xi_s) + \delta_{it}^5(\xi_s) \leqslant 1, \quad y_{it}^+(\xi_s) = M\mu_{it}^5(\xi_s), \quad d_{it}^-(\xi_s) = M\delta_{it}^5(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18o)}\\
& \mu_{it}^6(\xi_s) + \delta_{it}^6(\xi_s) \leqslant 1, \quad y_{it}^-(\xi_s) = M\mu_{it}^6(\xi_s), \quad d_{it}^+(\xi_s) = M\delta_{it}^6(\xi_s) && \forall i\in I, t\in T, s\in S && \text{(18p)}
\end{aligned}
$$

$$
P_t^{IN}(\xi_s) = -S\lambda_t(\xi_s) \tag{19}
$$

7.4 Replay

for each i:

$$
\begin{aligned}
\text{max} \quad & \sum_{t\in T}\left(P_t^{DA}x_t + \pi_s\left(P_t^{RT}(\xi_s)y_t^+(\xi_s) - P_t^{PN}(\xi_s)y_t^-(\xi_s) + P_t^{IN}(\xi_s)d_t^+(\xi_s) - P_t^{IN}(\xi_s)d_t^-(\xi_s)\right)\right) && && \text{(20a)}\\
\text{s.t.} \quad & R_t(\xi_s) - x_t = y_t^+(\xi_s) - y_t^-(\xi_s) + d_t^+(\xi_s) - d_t^-(\xi_s) + z_t^C(\xi_s) - z_t^D(\xi_s) && \forall t\in T, s\in S && \text{(20b)}\\
& R_t(\xi_s) + z_t^D(\xi_s) \geqslant y_t^+(\xi_s) + d_t^+(\xi_s) + z_t^C(\xi_s) && \forall t\in T, s\in S && \text{(20c)}\\
& z_{t+1}(\xi_s) = z_t(\xi_s) + \eta^C z_t^C(\xi_s) - \frac{1}{\eta^D} z_t^D(\xi_s) && \forall t\in T, s\in S && \text{(20d)}\\
& z_t^D(\xi_s) \leqslant \min\{\kappa^D, z_t(\xi_s)\} && \forall t\in T, s\in S && \text{(20e)}\\
& z_t^C(\xi_s) \leqslant \min\{\kappa^C, K_i - z_{tt}(\xi_s)\} && \forall t\in T, s\in S && \text{(20f)}\\
& 0 \leqslant z_t(\xi_s) \leqslant K_i && \forall t\in T, s\in S && \text{(20g)}\\
& z_t^C(\xi_s) \leqslant M'\phi_t^1(\xi_s), \quad z_t^D(\xi_s) \leqslant M'(1 - \phi_t^1(\xi_s)) && \forall t\in T, s\in S && \text{(20h)}\\
& y_t^-(\xi_s) \leqslant M\phi_t^2(\xi_s), \quad z_t^C(\xi_s) \leqslant M'(1 - \phi_t^2(\xi_s)) && \forall t\in T, s\in S && \text{(20i)}\\
& d_t^-(\xi_s) \leqslant M\phi_t^3(\xi_s), \quad z_t^C(\xi_s) \leqslant M'(1 - \phi_t^3(\xi_s)) && \forall t\in T, s\in S && \text{(20j)}\\
& d_t^+(\xi_s) \leqslant M\phi_t^4(\xi_s), \quad d_t^-(\xi_s) \leqslant M(1 - \phi_t^4(\xi_s)) && \forall t\in T, s\in S && \text{(20k)}\\
& y_t^+(\xi_s) \leqslant M\phi_t^5(\xi_s), \quad d_t^-(\xi_s) \leqslant M(1 - \phi_t^5(\xi_s)) && \forall t\in T, s\in S && \text{(20l)}\\
& y_t^-(\xi_s) \leqslant M\phi_t^6(\xi_s), \quad d_t^+(\xi_s) \leqslant M(1 - \phi_t^6(\xi_s)) && \forall t\in T, s\in S && \text{(20m)}
\end{aligned}
$$